

Conditional flow matching for genomic panel completion: inferring IMPACT505 alterations from the IMPACT341 targeted panel in non-small cell lung cancer

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1 Background

Comprehensive genomic profiling (CGP) guides treatment in NSCLC, but panel content, cost and turnaround time differ widely between institutions, and smaller targeted panels leave part of the tumour genome unmeasured.

EBAI class B. Under the ESMO basic requirements for AI-based biomarkers (EBAI), a model that predicts an established biomarker from an input different from the gold-standard test is class B; here the input is the smaller panel itself. Because the input differs from the reference test, risk and complexity exceed class A. Such tools may predict only a subset of alterations with sufficient sensitivity and specificity, and usually aim to **enrich** a population for confirmatory testing, or, where a negative result reliably excludes the alteration, to **rule out** testing. They do not substitute an established genomic test.

Objective. To learn the conditional distribution of alterations in genes absent from a targeted panel given the genes it measures, benchmark it against simple baselines, and evaluate it along the seven EBAI dimensions: comparator, performance, generalisability, fairness, explainability, cost and turnaround time.

2 Study design



Figure 1. Task definition. IMPACT341 ⊂ IMPACT505; the 164 genes present only in IMPACT505 are masked from the input and predicted.

Table 1. Cohort and target characteristics (MSK-CHORD).

Patients, NSCLC, sequenced on IMPACT505	2,226
Input genes (IMPACT341)	341
Target genes (IMPACT505 \ IMPACT341)	164
Altered target gene-patient pairs	1.21%
Patients with ≥1 altered target gene	[]%
Split (patient level): train / val / test	70 / 15 / 15% [confirm]

Alteration encoding. Target: one binary event per gene and patient (non-synonymous mutation, amplification or deep deletion; [state VUS and structural-variant handling]). Input: separate mutation, amplification and deletion indicators for the 341 assayed genes. One sample per patient. Genes not covered by a sample's panel are treated as missing, never as wild type.

3 Methods

3.1 Problem formulation. Given the observed IMPACT341 profile c , we learn the conditional distribution $p(x_1 \mid c)$ over the 164 target genes. Binary targets are embedded as $x_1 \in \{-1, +1\}^{164}$ (continuous relaxation); generated profiles are binarised by sign. Data were split by patient into 70% training, 15% validation (all model and hyper-parameter selection) and 15% test, evaluated once.

3.2 Conditional flow matching. A velocity field v_θ ([architecture]) is regressed onto the straight-line path between Gaussian noise and data:

$$x_t = (1 - t) x_0 + t x_1, \quad x_0 \sim \mathcal{N}(0, I), \quad t \sim \mathcal{U}(0, 1) \\ \mathcal{L}(\theta) = \mathbb{E} \parallel v_\theta(x_t, t \mid c) - (x_1 - x_0) \parallel^2$$

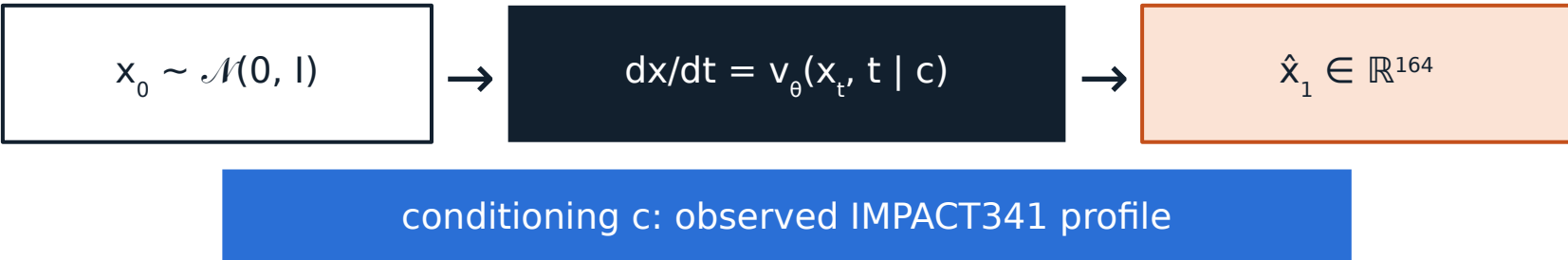


Figure 2. Sampling: the learned ODE, integrated from $t = 0$ to 1 ([solver, N steps]), transports noise to a completed 164-gene profile conditioned on the observed panel.

3.3 Probability read-out. At $t = 0$, x_t is independent of x_1 , so $v^*(x_0, 0 \mid c) = \mathbb{E}[x_1 \mid c] - x_0$ and $\hat{p} = \frac{1}{2}(1 + x_0 + v\theta(x_0, 0 \mid c))$, averaged over K noise draws. This avoids Monte Carlo ties at ~1% prevalence. [state read-out used]

3.4 Evaluation (EBAI). Pre-specified primary endpoint: pooled AUROC. Secondary: per-gene AUROC (genes with ≥5 test events; prevalence-only = 0.50); AUPRC (chance = 0.012); calibration plot, slope, intercept, O:E and Brier skill; at the intended-use threshold for ≥1 altered target gene, sensitivity, specificity, PPV, NPV and LR±; NRI over the burden + locus model and decision-curve analysis; joint fidelity (energy score versus independent Bernoulli draws). Comparators on identical splits: prevalence, burden + locus, L2 logistic regression. 95% CIs and paired differences by patient bootstrap. **Sample size (post hoc; EBAI asks a priori):** ~334 test patients give a 95% CI half-width of $\approx \pm 0.05$ for a patient-level AUROC of 0.79 (Hanley-McNeil, 43% prevalence assumed).

3.5 External replicate. MSK-IMPACT 2017 (cBioPortal, ODbL): IMPACT341 → 69 IMPACT410-only genes in 1,215 NSCLC patients; 3 × 5-fold patient-level cross-validation, AUROC computed within folds.

4 Results

4.1 Primary cohort. Across architectures and training paradigms, the flow model reached a pooled test AUROC of 0.77–0.79, above the prevalence-only baseline in every variant (Table 2).

Table 2. MSK-CHORD test set: AUROC pooled and per gene, and AUPRC (chance level = prevalence, 0.012); point estimate (95% CI).

Model	Pooled	Per gene	AUPRC
Prevalence only	[]	0.50	[]
Burden + locus	[]	[]	[]
L2 logistic regression	[]	[]	[]
Flow matching	0.77-0.79	[]	[]

Calibration and joint fidelity: Brier skill []; energy score, flow samples vs independent draws [] vs [].

4.2 Benchmark on public data. Pooled AUROC compares pairs across genes, so ranking genes by training frequency alone scores 0.715. Per gene, a 7-feature model matches 1,026-feature logistic regression (0.760 vs 0.728), and pan-cancer training adds +0.039 (95% CI 0.024–0.057; Figure 3).

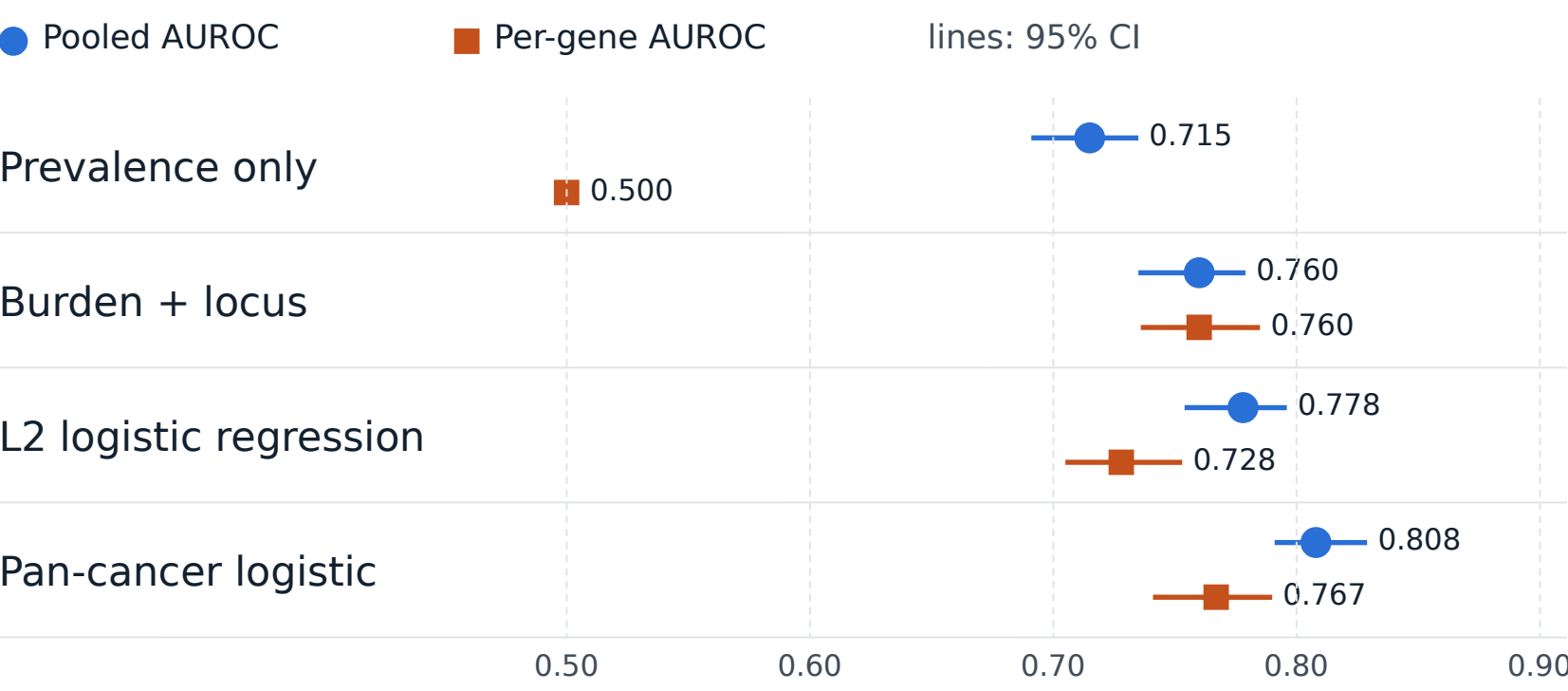


Figure 3. Replicate ($n = 1,215$): AUROC with patient-bootstrap 95% CI. Pan-cancer: trained on 7,773 patients with cancer type as covariate.

4.3 Simulated versus real input. In 167 patients sequenced on both panels, taking the input from the separate IMPACT341 assay instead of masking the IMPACT410 sample lowered pooled AUROC from 0.84 to 0.70 and per-gene AUROC from 0.80 to 0.61 (pan-cancer logistic regression). The paired samples also differ in specimen and time, so the gap combines assay shift and tumour heterogeneity.

4.4 Sources of signal. Amplifications were highly discriminable (AUROC 0.95), explained largely by co-amplification with neighbouring assayed genes (NFKBIA with NKX2-1 in 39/39 cases). Mutations reached 0.73, no better than three mutational-burden counts alone (Figure 4).

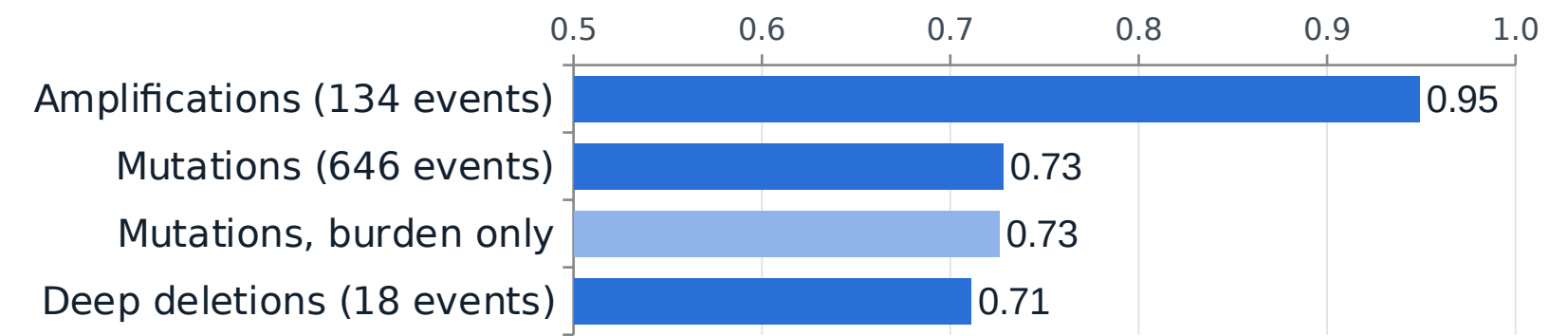


Figure 4. Replicate: per-gene AUROC by event type (positives-weighted mean; axis origin 0.50), pan-cancer logistic regression.

5 EBAI assessment

EBAI class B · indirect measure of existing biomarkers · use: enrichment			
Essential	Comparator	Met	IMPACT505 NGS on the same sample
	Discrimination	Met	Pooled AUROC 0.77–0.79; per gene []
	Operating point	Partial	Replicate: Se 0.90, NPV 0.86, LR– 0.22
	Calibration	Partial	Replicate: slope 0.97, O:E 1.08; plot []
	Indep. cohort	Not yet	Single institution; external cohort needed
Recommended	Assay variation	Partial	Real IMPACT341 input: AUROC 0.84 → 0.70
	Fairness	Not yet	Sex, ancestry, smoking, histology
	Explainability	Met	Ablation: CN locus + mutational burden
Additional	Cost	Partial	Reuses routine panel data; cost analysis []
	Turnaround	Partial	Inference in seconds; end-to-end time []
Validation	Sample size	Partial	Post hoc: $\approx \pm 0.05$ AUROC with 334 test patients
	NRI / DCA	Not yet	Versus burden + locus model; triage threshold
Post-adoption	Monitoring	Not yet	Drift with panel versions and populations

Figure 5. Assessment along the seven EBAI dimensions and validation requirements for class B. Replicate values from MSK-IMPACT 2017; [] to compute on MSK-CHORD. CN, copy number; DCA, decision-curve analysis; NRI, net reclassification improvement.

6 Discussion

- A targeted panel predicts part of the unassayed genome; per-gene AUROC and simple baselines, not pooled AUROC alone, show how much.
- The signal is copy-number co-location and mutational burden; a generative model must add joint fidelity to justify its complexity.
- EBAI class B: NPV 0.86 at 90% sensitivity does not support rule-out; use is enrichment, with confirmatory CGP.
- Next: independent multi-centre validation, fairness audit, DCA and post-deployment drift monitoring.

References

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Data and code

MSK-CHORD and msk_impact_2017 via cBioPortal. Code: [repository link]

Disclosures · Contact

[Conflict-of-interest statement]
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